

LAFAYETTE HILL, PA, USA

WMO: 720304

Lat: 40.100N

Lon: 75.267W

Elev: 92

StdP: 100.22

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 64752

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.2	-8.8	-20.2	0.6	-8.8	-17.5	0.8	-6.8	8.4	3.9	7.5	2.0	1.8	300	0.420

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	32.7	23.3	31.4	22.8	30.2	22.1	25.1	30.2	24.3	29.1	23.6	28.0	2.9	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.6	18.7	27.4	22.8	17.7	26.4	22.4	17.3	25.9	76.9	29.8	73.5	29.2	70.9	28.2	27.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.4	6.2	5.3	-14.7	35.4	2.6	2.1	-16.5	36.9	-18.1	38.1	-19.5	39.3	-21.4	40.8		
			-15.4	26.8	2.6	0.6	-17.3	27.2	-18.8	27.6	-20.2	27.9	-22.1	28.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.1	-0.1	1.0	5.3	11.4	16.7	21.4	24.0	22.8	19.2	12.8	6.9	2.9
	DBStd	9.44	5.33	4.96	5.08	4.60	4.47	3.42	2.80	2.86	3.70	4.55	4.42	4.88
	HDD10.0	1133	314	256	163	34	3	0	0	0	0	23	114	227
	HDD18.3	2811	571	486	405	214	87	12	1	2	34	179	343	478
	CDD10.0	1896	2	3	17	76	211	342	434	398	276	111	20	7
	CDD18.3	532	0	0	0	6	36	104	176	141	61	8	0	0
	CDH23.3	4933	0	0	6	82	366	955	1694	1251	521	58	1	0
	CDH26.7	1621	0	0	0	19	98	301	653	398	144	8	0	0
Wind	WSAvg	1.9	2.2	2.3	2.5	2.5	2.0	1.8	1.4	1.3	1.4	1.7	1.9	2.1
Precipitation	PrecAvg	1185	91	66	108	96	92	102	114	109	115	107	82	105
	PrecMax	1591	155	146	175	250	194	271	257	239	299	233	134	237
	PrecMin	883	37	16	28	31	31	37	28	37	7	8	27	28
	PrecStd	175	34	32	40	47	36	55	52	47	70	62	30	47
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.2	19.0	23.9	29.8	31.4	33.9	35.9	34.0	32.4	27.5	22.0	19.1
		MCWB	13.9	14.3	14.9	18.1	21.8	22.9	24.0	22.3	23.1	20.3	16.4	16.9
	2%	DB	13.8	15.1	20.0	26.0	29.5	32.0	32.9	32.1	30.2	25.0	19.1	16.1
		MCWB	11.3	11.0	13.3	16.3	20.7	22.6	23.9	23.4	21.9	19.1	15.2	14.4
	5%	DB	11.0	12.0	16.3	22.7	27.4	30.0	32.0	30.2	27.9	22.6	17.2	12.8
		MCWB	8.8	8.3	10.9	14.7	19.6	21.8	23.5	22.2	20.6	18.2	13.2	10.9
	10%	DB	7.7	7.9	13.5	20.3	25.2	27.9	30.3	28.8	26.4	21.0	14.9	10.8
		MCWB	5.5	5.0	9.4	13.7	18.6	20.5	22.5	21.7	20.0	17.1	11.4	8.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.8	15.7	16.9	19.6	23.5	25.0	26.7	26.1	25.2	22.5	18.4	17.4
		MCDB	16.6	18.0	21.6	24.6	29.2	31.2	33.0	30.4	29.6	25.5	19.8	18.3
	2%	WB	12.0	11.9	14.4	18.0	22.4	23.9	25.4	24.9	23.9	21.1	15.8	14.7
		MCDB	12.9	13.9	18.3	23.7	27.6	29.3	30.7	29.3	28.2	23.3	18.1	16.3
	5%	WB	8.5	8.0	12.0	16.5	21.1	22.9	24.5	24.0	22.9	19.6	14.0	11.8
		MCDB	10.9	10.8	15.5	21.3	25.1	28.0	29.5	27.9	26.3	22.1	16.7	12.8
	10%	WB	5.8	5.6	9.7	14.7	19.7	22.1	23.7	23.1	21.9	17.7	12.2	8.5
		MCDB	7.9	8.0	13.3	19.1	23.9	26.5	28.3	26.6	24.9	19.4	14.1	10.0
Mean Daily Temperature Range	5% DB	MDBR	8.2	8.6	9.6	11.6	11.4	10.9	10.8	10.7	11.4	10.9	10.0	7.8
		MCDBR	12.4	13.3	14.4	15.5	13.9	12.9	12.3	12.0	13.2	13.3	13.3	11.1
		MCWBR	10.4	10.1	9.2	7.7	6.5	5.2	4.4	4.9	6.1	7.7	9.3	9.7
	5% WB	MCDBR	11.5	12.5	12.8	12.6	12.1	11.1	10.3	9.5	10.7	10.1	11.4	10.3
		MCWBR	10.3	10.5	9.6	8.2	6.6	5.2	4.4	4.6	5.9	7.1	9.4	10.1
Clear-Sky Solar Irradiance	taub	0.309	0.320	0.360	0.391	0.437	0.497	0.498	0.492	0.405	0.358	0.325	0.307	
	taud	2.455	2.412	2.368	2.308	2.208	2.083	2.094	2.097	2.310	2.453	2.511	2.511	
	Ebn at Noon	856	898	892	883	845	793	787	780	833	842	832	832	
	Edh at Noon	79	95	111	126	142	161	158	154	116	91	75	70	
All-Sky Solar Radiation	RadAvg	1.90	2.80	3.70	4.80	5.35	5.85	5.97	5.19	4.30	3.06	2.17	1.64	
	RadStd	0.16	0.31	0.29	0.43	0.64	0.47	0.29	0.33	0.49	0.32	0.19	0.17	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (75 neighbors)	+0.41	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+121	+68

Nomenclature: See separate page